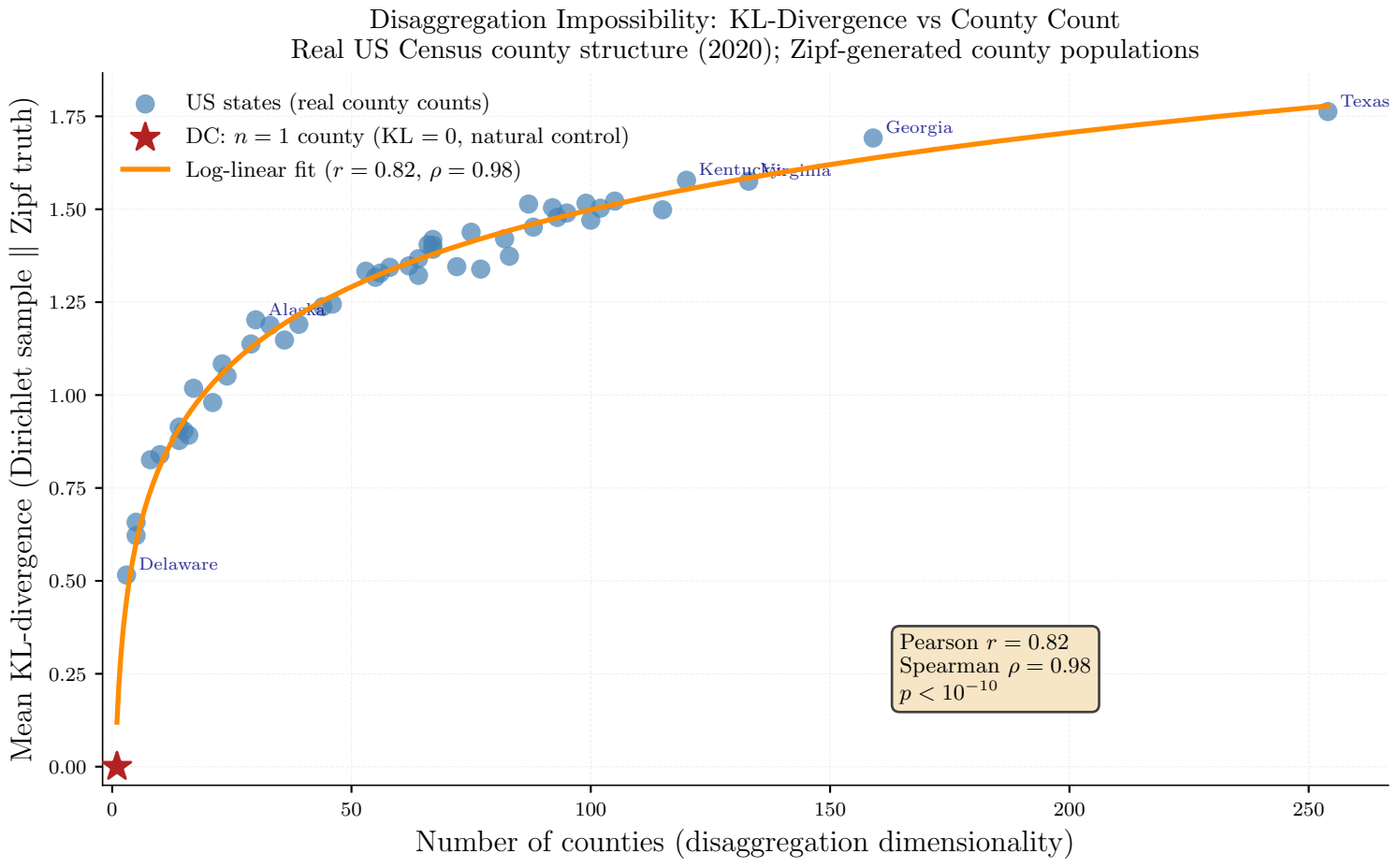
Disaggregation Impossibility: KL-Divergence vs County Count
Real US Census county structure (2020); Zipf-generated county populations
US states (real county counts)
DC: $n = 1$ county (KL = 0, natural control)
Log-linear fit ($r = 0.82$, $\rho = 0.98$)
Texas
Georgia
Kentucky
Virginia
Alaska
Delaware
Pearson $r = 0.82$
Spearman $\rho = 0.98$
$p < 10^{-10}$
Mean KL-divergence (Dirichlet sample || Zipf truth)
Number of counties (disaggregation dimensionality)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
0
50
100
150
200
250